Supplementary Material for Target-Network Refresh via Adaptive Decay Estimation in Deep Reinforcement Learning

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Appendix A
Proofs for the Stability–Efficiency Analysis

This appendix provides detailed derivations for the stability–efficiency analysis in the main text. Section A-1 establishes the PL-based local target-fitting contraction used in Assumption 2. The subsequent proofs derive the outer-loop Bellman error bound in Theorem 1, the unique-minimizer property of the step-wise convergence rate in Proposition 1, and the oracle advantage of adaptive scheduling in Proposition 2. Together, these results formalize two requirements underlying TRADE: refresh must be delayed sufficiently to preserve contraction, while its timing must adapt to local fitting dynamics to avoid redundant optimization after saturation.

A-1. Derivation of Local Target-Fitting Contraction

This section derives the contraction rate used in Assumption 2. When the target network is frozen and the local target-fitting objective is μ-PL and β-smooth, gradient descent reduces the distance to the frozen Bellman target geometrically. The derivation first establishes a local gradient-descent contraction factor and then expresses the resulting progress on the environment-step clock used in the outer-loop analysis.

Assumption 2. During target-update cycle m, the target $Q_{\theta_m^-}$ is fixed. If the local target-fitting objective is μ-PL, β-smooth, and locally realizable, then gradient descent with $\alpha < 2/\beta$ yields the single-update factor

$$\bar{\rho}_m = \sqrt{1 - 2\alpha\mu\left(1 - \frac{\alpha\beta}{2}\right)} \in (0,1).$$

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Writing $\rho_m \in (0,1)$ for the effective local contraction factor per environment interaction, after k environment steps,

$$\|Q_{\theta_{m,k}} - \mathcal{T}Q_{\theta_m^-}\| \le \rho_m^k \|Q_{\theta_{m,0}} - \mathcal{T}Q_{\theta_m^-}\|.$$

Proof. During the m-th target-update cycle, the target parameters θ_m^- are fixed. Consider the idealized fitting objective

$$L_m(\theta) = \frac{1}{2}\|Q_\theta - \mathcal{T}Q_{\theta_m^-}\|^2. \tag{1}$$

Assume that L_m is μ-PL and β-smooth in the local region visited by gradient descent, and assume exact realizability so that there exists θ_m^* satisfying

$$Q_{\theta_m^*} = \mathcal{T}Q_{\theta_m^-}, \qquad L_m(\theta_m^*) = 0. \tag{2}$$

Using u to index gradient updates, consider

$$\theta_{u+1} = \theta_u - \alpha\nabla L_m(\theta_u), \tag{3}$$

β-smoothness gives

$$\begin{aligned} L_m(\theta_{u+1}) &\le L_m(\theta_u) + \nabla L_m(\theta_u)^\top(\theta_{u+1} - \theta_u) \\ &\quad + \frac{\beta}{2}\|\theta_{u+1} - \theta_u\|^2 \\ &= L_m(\theta_u) - \alpha\|\nabla L_m(\theta_u)\|^2 + \frac{\beta\alpha^2}{2}\|\nabla L_m(\theta_u)\|^2 \\ &= L_m(\theta_u) - \alpha\left(1 - \frac{\alpha\beta}{2}\right)\|\nabla L_m(\theta_u)\|^2. \end{aligned} \tag{4}$$

Applying the PL condition,

$$\|\nabla L_m(\theta_u)\|^2 \ge 2\mu\left(L_m(\theta_u) - L_m(\theta_m^*)\right) = 2\mu L_m(\theta_u), \tag{5}$$

we obtain

$$L_m(\theta_{u+1}) \le \left[1 - 2\alpha\mu\left(1 - \frac{\alpha\beta}{2}\right)\right] L_m(\theta_u). \tag{6}$$

Let

$$\bar{\rho}_m = \sqrt{1 - 2\alpha\mu\left(1 - \frac{\alpha\beta}{2}\right)}. \tag{7}$$

When $\alpha < 2/\beta$ and the resulting $\bar{\rho}_m \in (0,1)$, taking square roots and unrolling over local gradient updates yields geometric contraction. On the environment-step clock used by the target scheduler, we summarize the aggregate fitting progress within the m-th cycle by an effective factor $\rho_m \in (0,1)$. Therefore, after k environment steps,

$$\|Q_{\theta_{m,k}} - \mathcal{T}Q_{\theta_m^-}\| \le \rho_m^k \|Q_{\theta_{m,0}} - \mathcal{T}Q_{\theta_m^-}\|. \tag{8}$$

Thus, the PL-based local target-fitting contraction in Assumption 2 holds on the common interaction clock. □

A-2. Proof of the Outer-loop Error Bound

Theorem 1. Under Assumptions 1 and 2, after C environment steps in target-update cycle m, the outer-loop error satisfies

$$\delta_{m+1} \leq \left[\gamma + (1+\gamma)\rho_m^C\right] \delta_m.$$

Proof. By definition,

$$\delta_{m+1} = \|Q_{\theta_{m,C}} - Q^*\|. \tag{9}$$

Using the fixed-point relation $Q^* = \mathcal{T}Q^*$ and the triangle inequality,

$$\begin{aligned} \delta_{m+1} &= \|Q_{\theta_{m,C}} - \mathcal{T}Q^*\| \\ &\leq \|Q_{\theta_{m,C}} - \mathcal{T}Q_{\theta_m^-}\| + \|\mathcal{T}Q_{\theta_m^-} - \mathcal{T}Q^*\|. \end{aligned} \tag{10}$$

By Assumption 1,

$$\|\mathcal{T}Q_{\theta_m^-} - \mathcal{T}Q^*\| \leq \gamma \|Q_{\theta_m^-} - Q^*\| = \gamma\delta_m. \tag{11}$$

By Assumption 2,

$$\|Q_{\theta_{m,C}} - \mathcal{T}Q_{\theta_m^-}\| \leq \rho_m^C \|Q_{\theta_{m,0}} - \mathcal{T}Q_{\theta_m^-}\|. \tag{12}$$

Since $\theta_{m,0} = \theta_m^-$,

$$\begin{aligned} \|Q_{\theta_{m,0}} - \mathcal{T}Q_{\theta_m^-}\| &= \|Q_{\theta_m^-} - \mathcal{T}Q_{\theta_m^-}\| \\ &= \|Q_{\theta_m^-} - Q^* + \mathcal{T}Q^* - \mathcal{T}Q_{\theta_m^-}\| \\ &\leq \|Q_{\theta_m^-} - Q^*\| + \|\mathcal{T}Q^* - \mathcal{T}Q_{\theta_m^-}\| \\ &\leq (1+\gamma)\delta_m. \end{aligned} \tag{13}$$

Combining the above inequalities gives

$$\delta_{m+1} \leq \left[\gamma + (1+\gamma)\rho_m^C\right] \delta_m. \tag{14}$$

□

A-3. Proof of Proposition 1: Unique Minimizer

Proposition 1. Let

$$\mathcal{R}_m(C) = \frac{\ln\big(\gamma + (1+\gamma)\rho_m^C\big)}{C}$$

be the local step-wise convergence rate on the contractive domain $C \in (C_{\mathrm{lb}}, \infty)$. Then $\mathcal{R}_m(C)$ possesses a unique global minimizer $C_m^* \in (C_{\mathrm{lb}}, \infty)$.

Proof. Let $u = C \ln \rho_m$. Since $\rho_m \in (0,1)$ and $\ln \rho_m < 0$, the mapping $C \mapsto u$ is strictly decreasing. Let $T = \frac{\ln((1-\gamma)/(1+\gamma))}{\ln \rho_m}$ be the continuous threshold and $u_T = T \ln \rho_m$, so that $\gamma + (1+\gamma)e^{u_T} = 1$. Since $C_{\mathrm{lb}} = \lceil T \rceil \geq T$, the domain $C > C_{\mathrm{lb}}$ corresponds to $u \in (-\infty, u_T)$ (a superset of $(-\infty, u_{\mathrm{lb}})$). Substituting $\rho_m^C = e^u$ yields

$$\mathcal{R}_m(C) = \ln \rho_m \cdot \frac{f(u)}{u}, \qquad f(u) = \ln\big(\gamma + (1+\gamma)e^u\big). \tag{15}$$

Since $\ln \rho_m < 0$ is a constant, minimizing $\mathcal{R}_m$ over C is equivalent to maximizing $\tilde{R}(u) = f(u)/u$ over $u \in (-\infty, u_T)$. The derivative of $\tilde{R}$ is

$$\tilde{R}'(u) = \frac{g(u)}{u^2}, \qquad g(u) = u f'(u) - f(u). \tag{16}$$

Since $u^2 > 0$, the sign of $\tilde{R}'$ is determined by g. Differentiating,

$$g'(u) = u f''(u), \tag{17}$$

where $f''(u) = \frac{(1+\gamma)\gamma e^u}{(\gamma + (1+\gamma)e^u)^2} > 0$, so f is strictly convex. On the domain $u < 0$, we have $g'(u) = u f''(u) < 0$, hence g is strictly decreasing.

Evaluating g at the two boundaries yields the required sign change. As $u \to -\infty$, $f'(u) \to 0$ exponentially and $f(u) \to \ln \gamma$, giving $g(u) \to -\ln \gamma > 0$. At the right boundary, $f(u_T) = 0$ (since $\gamma + (1+\gamma)e^{u_T} = 1$) and $f'(u_T) = (1-\gamma)$, so $g(u_T) = u_T(1-\gamma) < 0$.

Since g is continuous and strictly decreasing from a positive limit to a negative value, the equation $g(u) = 0$ has exactly one solution $u^* \in (-\infty, u_T)$. Consequently, $\tilde{R}$ increases on $(-\infty, u^*)$ and decreases on (u^*, u_T), yielding a unique maximum. Because $\ln \rho_m < 0$, $\mathcal{R}_m$ has a unique global minimum on $(C_{\mathrm{lb}}, \infty)$. □

A-4. Proof of the Oracle Advantage

Proposition 2. Let T be the total number of environment-interaction steps, let $m(t)$ denote the local target-fitting regime at step t, and let $\mathcal{C} = \{C \in \mathbb{N} : C \geq C_{\min}\}$. Define

$$C_{m(t)}^* \in \arg\min_{C \in \mathcal{C}} \mathcal{R}_{m(t)}(C).$$

For any constant $C_{\mathrm{fix}} \in \mathcal{C}$, define

$$\begin{aligned} r_t^{\mathrm{adapt}} &= \mathcal{R}_{m(t)}(C_{m(t)}^*), \\ r_t^{\mathrm{fix}} &= \mathcal{R}_{m(t)}(C_{\mathrm{fix}}). \end{aligned}$$

Under the same sequence of local rate functions,

$$\sum_{t=1}^{T} r_t^{\mathrm{adapt}} \leq \sum_{t=1}^{T} r_t^{\mathrm{fix}}.$$

The inequality is strict if there exists at least one step t for which $C_{\mathrm{fix}} \notin \arg\min_{C \in \mathcal{C}} \mathcal{R}_{m(t)}(C)$.

Proof. Let T be the total number of environment-interaction steps. For each step t, let $m(t)$ denote the local target-fitting regime associated with that step. The oracle adaptive interval is defined by

$$C_{m(t)}^* \in \arg\min_{C \in \mathcal{C}} \mathcal{R}_{m(t)}(C). \tag{18}$$

Therefore, for any fixed interval $C_{\mathrm{fix}} \in \mathcal{C}$, we have

$$\mathcal{R}_{m(t)}(C_{m(t)}^*) \leq \mathcal{R}_{m(t)}(C_{\mathrm{fix}}) \tag{19}$$

for every environment-interaction step $t = 1, \ldots, T$.

Define

$$r_t^{\mathrm{adapt}} = \mathcal{R}_{m(t)}(C_{m(t)}^*), \qquad r_t^{\mathrm{fix}} = \mathcal{R}_{m(t)}(C_{\mathrm{fix}}). \tag{20}$$

Summing the pointwise inequality over the same environment-interaction horizon gives

$$\sum_{t=1}^{T} r_t^{\mathrm{adapt}} \leq \sum_{t=1}^{T} r_t^{\mathrm{fix}}. \tag{21}$$

If there exists at least one step t such that $C_{\mathrm{fix}} \notin \arg\min_{C \in \mathcal{C}} \mathcal{R}_{m(t)}(C)$, then the inequality is strict for that step, and hence the accumulated inequality is also strict.

Equivalently, when the accumulated error bound is written on the logarithmic scale as

$$\ln \frac{\delta_T^{\pi}}{\delta_0} \leq \sum_{t=1}^{T} r_t^{\pi}, \tag{22}$$

the adaptive oracle schedule yields a no larger, and in the nonstationary case strictly smaller, final error bound than any fixed interval schedule under the same environment-interaction budget. □

Appendix B
Additional Experimental Figures and Tables

This appendix groups supplementary experiments by benchmark suite, pairing representative environment snapshots with hyper-parameters and additional learning curves. Within each suite, the visual context and shared settings precede the corresponding results. Atari covers value-based visual control, DMC and Meta-World cover continuous control and manipulation, and MuJoCo adds high-dimensional locomotion diagnostics.

B-1. Cliff Walking

Figure 1 reports the complete Cliff Walking parameter search. The main paper retains $C \in \{1, 100, 500\}$ in Figure 1 to illustrate short, intermediate, and long target-refresh regimes. The full search confirms that both learning and target-value convergence are sensitive to this interval.

B-2. Atari 2600

1) Environment and Parameters: Figure 2 shows four representative Atari environments with distinct visual settings and scoring dynamics, illustrating the diversity of the value-based DDQN+TRADE benchmark.

The complete 30-environment Atari comparison is reported in main-text Table I. Table I summarizes the corresponding DDQN hyper-parameters. DDQN and DDQN+TRADE use identical training budgets and evaluation protocols, thereby isolating the effect of target-update scheduling.

2) Additional Results: Figure 3 provides full-range critic losses corresponding to the limited-range panels in Fig. 7. The retained early transients and rare spikes aid instability diagnosis but would compress the main operating range.

B-3. DeepMind Control Suite

1) Environment and Parameters: Figure 5 shows four representative DMC environments. These examples include object-centric control, articulated manipulation, high-dimensional locomotion, and reaching, matching the range of dynamics covered by the DMC experiments.

Table II reports the shared continuous-control hyper-parameters for DMC and Meta-World. The additional DMC learning curves extend the main-text evaluation to further tasks and algorithm integrations, including the complete BRO comparison.

2) Additional Results: Figure 4 extends the SAC evaluation on DMC beyond the three representative tasks shown in the main text. It covers additional balance, locomotion, manipulation, and swimmer-style control tasks, providing a broader view of how TRADE affects SAC target updates across heterogeneous continuous-control dynamics.

Figure 6 provides additional REDQ comparisons on DMC tasks. These curves test whether TRADE remains useful when added to REDQ's ensemble-critic training pipeline.

Figure 7 evaluates TRADE within MR.Q on DMC, testing compatibility with its representation-learning pipeline.

Figure 8 consolidates the complete BRO evaluation on DMC. The *cheetah_run*, *hopper_hop*, and *walker_walk* comparisons evaluate the tasks used in the principal benchmark grid, while *fish_swim* and *pendulum_swingup* broaden the evaluation to additional dynamics.

B-4. Meta-World

1) Environment and Parameters: Figure 9 shows the four Meta-World environment snapshots available in our environment-visualization set. These manipulation tasks emphasize sparse or delayed success signals, making them useful for evaluating whether adaptive target refresh improves success-rate learning.

The Meta-World experiments use the shared continuous-control configuration reported in Table II, with task-dependent horizons and success-rate evaluation. This parameter table is placed with the DMC setup because the two continuous-control suites share the same optimizer, replay, batch-size, and TRADE tolerance settings.

2) Additional Results: Figure 10 reports additional SAC comparisons on Meta-World manipulation tasks together with the *button-press-topdown* tolerance analysis. The learning curves evaluate adaptive refresh under success-rate metrics, while the sensitivity panel demonstrates robustness across positive tolerances and attains the highest mean success at $\tau = 10^{-3}$.

Figure 11 reports the complete BRO evaluation on Meta-World. These comparisons evaluate TRADE under sparse manipulation success signals while preserving BRO's regularization, optimism, and replay design.

B-5. MuJoCo

1) Environment and Parameters: Figure 12 shows the four MuJoCo locomotion environments used in our supplementary continuous-control experiments: *Ant*, *HalfCheetah*, *Hopper*, and *Walker2d*. These tasks provide additional high-dimensional locomotion tests beyond DMC.

The MuJoCo runs follow the REDQ and MR.Q training protocols used in the main experiments, with TRADE changing only target-refresh timing. The corresponding figures therefore isolate the effect of adaptive scheduling under high-dimensional locomotion dynamics and stronger continuous-control baselines.

2) Additional Results: Figure 13 reports the MuJoCo component of the REDQ experiments. These tasks evaluate TRADE with ensemble critics on standard locomotion benchmarks.

Figure 14 reports the MuJoCo component of the MR.Q integration. The purpose is to verify that TRADE can be inserted into a stronger modern baseline without changing the surrounding representation-learning and regularization components.

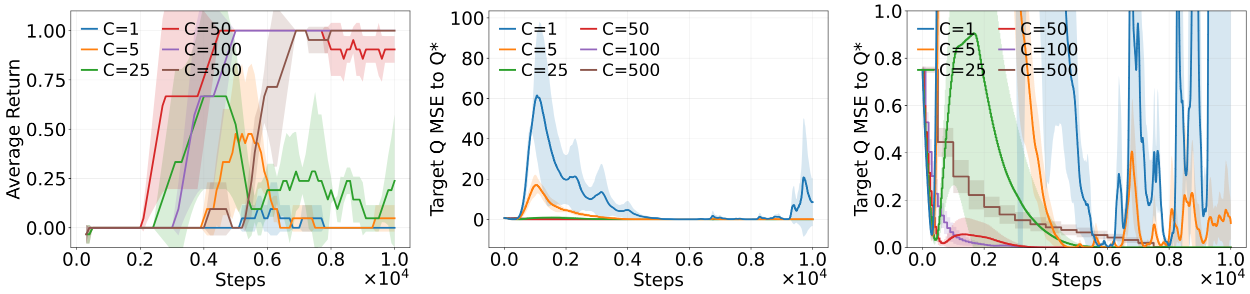
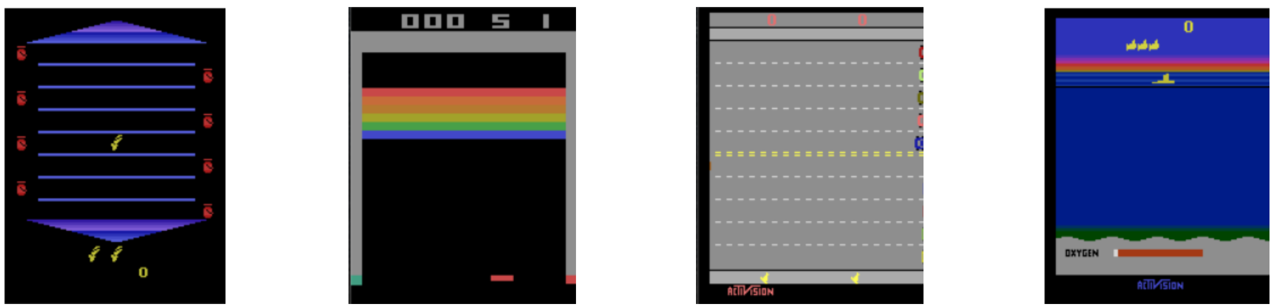


Fig. 1. Complete Cliff Walking parameter-search results for the tested fixed target-update intervals. Both very short and very long intervals degrade learning or delay target-value convergence, confirming the need to adapt the refresh interval to the fitting dynamics.

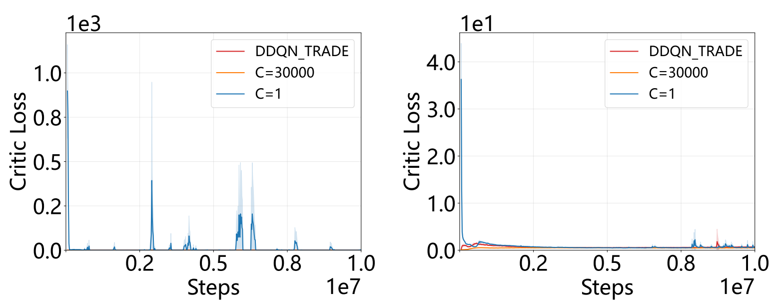


(a) *Asterix* (b) *Breakout* (c) *Freeway* (d) *Seaquest*

Fig. 2. Representative Atari 2600 environments used in the discrete visual-control evaluation. Their distinct visual layouts and scoring dynamics provide a heterogeneous test of whether TRADE transfers across value-learning tasks.

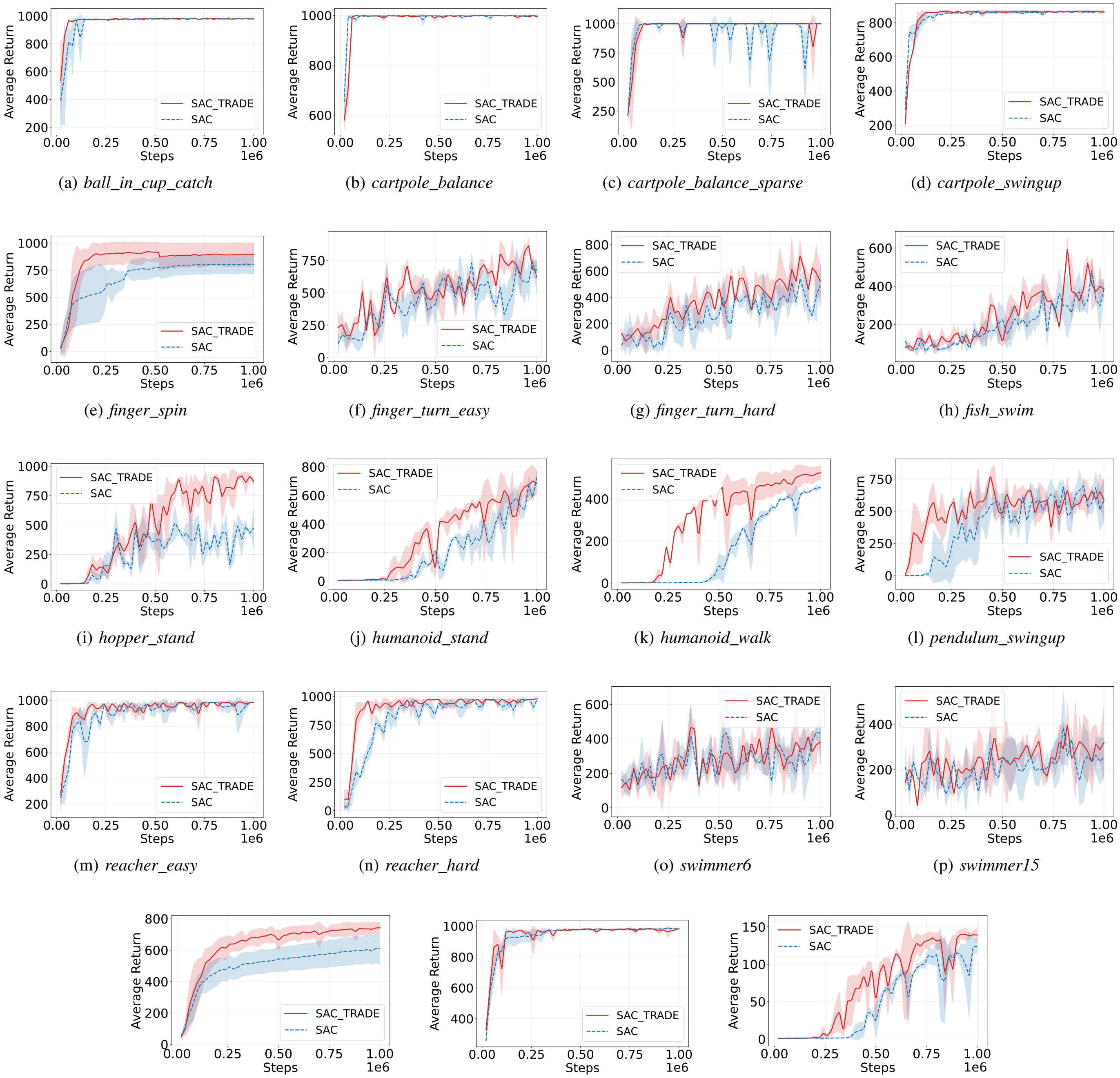
TABLE I
HYPER-PARAMETERS OF EXPERIMENTS ON ATARI ENVIRONMENTS.

Hyper-parameter	**Value**	**Hyper-parameter**	**Value**
Total environment steps	10M	Initial exploration ϵ	1.0
Replay buffer size	1.0×10^5	Final exploration ϵ	0.1
Mini-batch size	32	Exploration decay steps	100K
Optimizer	RMSprop	Exploration schedule	linear decay
Learning rate	2.5×10^{-4}	Warm-up steps	50K
Discount factor γ	0.99	Fixed target update interval	30K
Training frequency	every 4 steps	TRADE tolerance τ	1.0×10^{-3}



(a) *Frostbite*: full-range critic loss (b) *MsPacman*: full-range critic loss

Fig. 3. Full-range critic-loss diagnostics for DDQN+TRADE and fixed hard-update DDQN baselines on Atari. Compared with the limited-range panels in Fig. 7, these plots expose the early transients and extreme spikes, especially under the overly frequent $C = 1$ target-update schedule.



(a) *ball_in_cup_catch* (b) *cartpole_balance* (c) *cartpole_balance_sparse* (d) *cartpole_swingup*

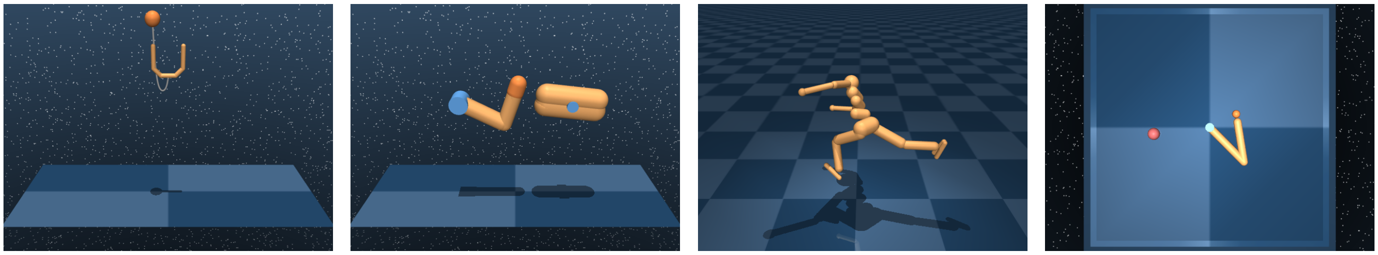
(e) *finger_spin* (f) *finger_turn_easy* (g) *finger_turn_hard* (h) *fish_swim*

(i) *hopper_stand* (j) *humanoid_stand* (k) *humanoid_walk* (l) *pendulum_swingup*

(m) *reacher_easy* (n) *reacher_hard* (o) *swimmer6* (p) *swimmer15*

(q) *walker_run* (r) *walker_stand* (s) *humanoid_run*

Fig. 4. Additional SAC comparisons on DeepMind Control Suite tasks; the main-grid environments *cheetah_run*, *hopper_hop*, and *walker_walk* appear in Fig. 3. Across this broader task set, SAC+TRADE generally matches or improves the baseline and often learns faster, supporting transfer across diverse continuous-control dynamics.

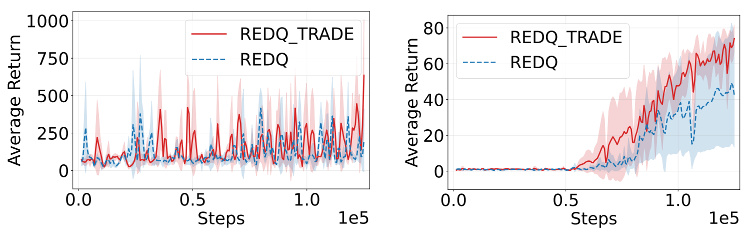


(a) *ball_in_cup_catch* (b) *finger_spin* (c) *humanoid_run* (d) *reacher_hard*

Fig. 5. Representative DeepMind Control Suite environments used in the continuous-control evaluation. They span object-centric control, articulated manipulation, high-dimensional locomotion, and reaching, testing TRADE across heterogeneous dynamics.

TABLE II
HYPER-PARAMETERS OF EXPERIMENTS ON DMC AND META-WORLD ENVIRONMENTS.

Hyper-parameter	**DMC**	**Meta-World**
Total environment steps	1M	1M–3M (task-dependent)
Replay buffer size	1.0×10^6	1.0×10^6
Mini-batch size	256	256
Discount factor γ	0.99	0.99
SAC entropy temperature	auto-tuning, initial $\alpha = 0.1$	auto-tuning, initial $\alpha = 1.0$
Random action warm-up	1K steps	10K steps
Episode horizon	1000 steps	500 steps
Evaluation interval	every 20K steps	every 10K steps
TRADE tolerance τ	1.0×10^{-3}	1.0×10^{-3}
Evaluation protocol	deterministic policy	deterministic policy + success metric



(a) *fish_swim* (b) *humanoid_run*

Fig. 6. Additional REDQ comparisons on DMC tasks. TRADE remains comparable on the noisy *fish_swim* task and improves learning on *humanoid_run*, showing that adaptive refresh remains useful within REDQ's ensemble-critic pipeline.

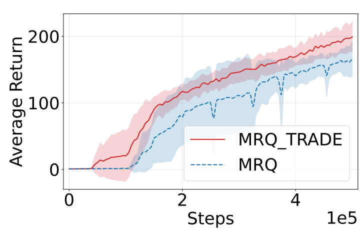
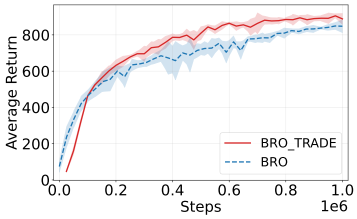
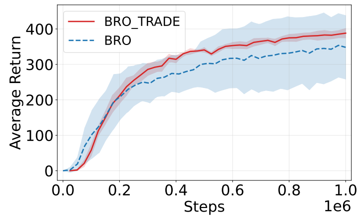


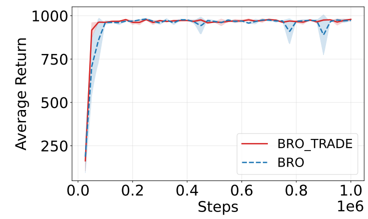
Fig. 7. Additional MR.Q comparison on the DMC task *humanoid_run*. MR.Q+TRADE learns earlier and reaches a higher mean return, indicating compatibility with MR.Q's representation-learning pipeline.



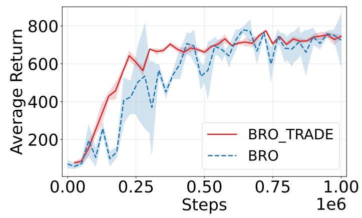
(a) *cheetah_run*



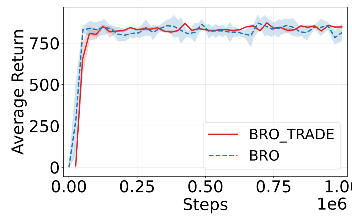
(b) *hopper_hop*



(c) *walker_walk*

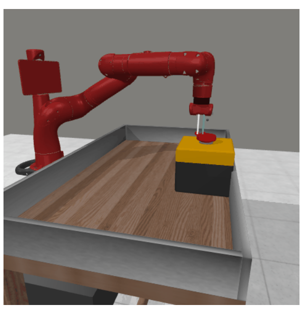


(d) *fish_swim*

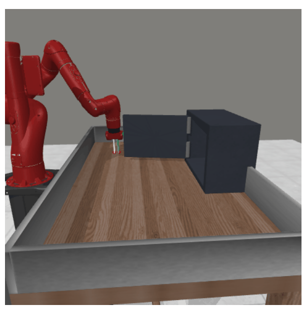


(e) *pendulum_swingup*

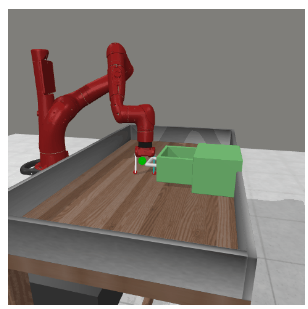
Fig. 8. BRO comparisons on DMC. TRADE improves or preserves the strong BRO baseline across the three representative benchmark tasks and the two additional control tasks. It accelerates early learning on *fish_swim*, while the near-ceiling *walker_walk* and *pendulum_swingup* results show that adaptive refresh does not sacrifice saturated performance.



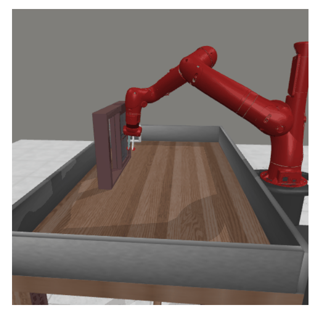
(a) *button-press-topdown*



(b) *door-open*

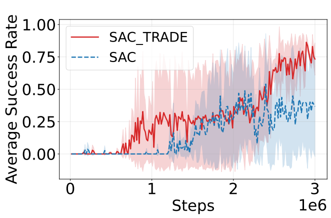


(c) *drawer-open*

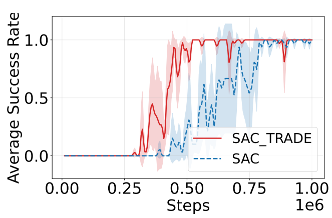


(d) *window-open*

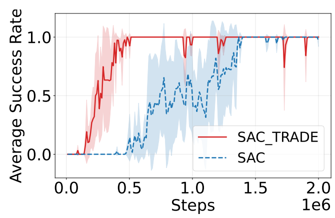
Fig. 9. Representative Meta-World manipulation environments. These tasks combine distinct object interactions with sparse or delayed success signals, testing whether TRADE generalizes across robotic manipulation objectives.



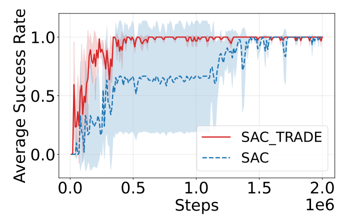
(a) *box-close*



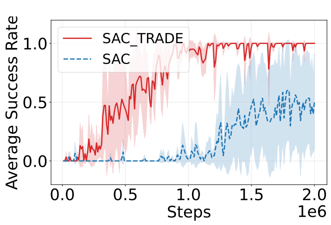
(b) *door-open*



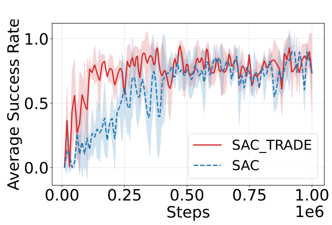
(c) *drawer-open*



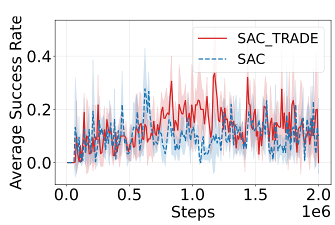
(d) *faucet-open*



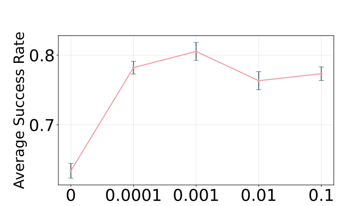
(e) *handle-pull*



(f) *plate-slide*

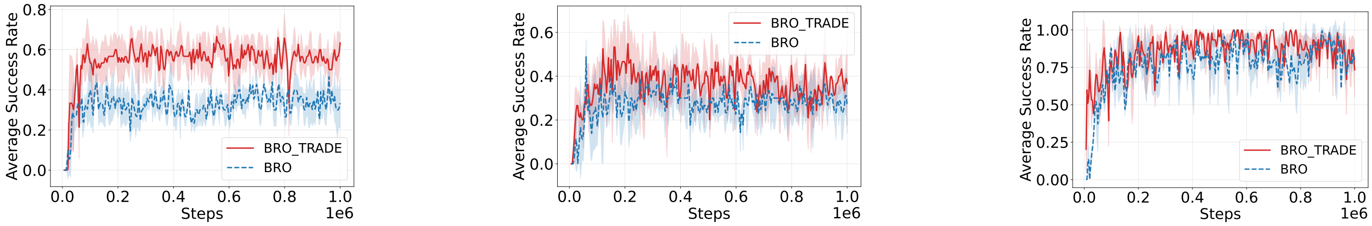


(g) *reach*



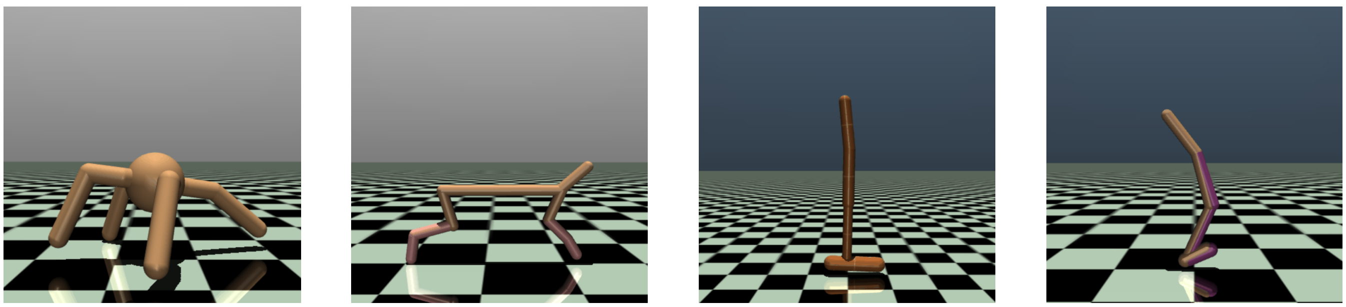
(h) τ sensitivity

Fig. 10. Additional Meta-World results. Panels (a)–(g) compare SAC+TRADE with SAC; the main-grid environments *button-press-topdown*, *dial-turn*, and *window-open* appear in Fig. 4. Panel (h) reports sensitivity to τ on *button-press-topdown* over 5×10^5 environment steps, with markers and error bars denoting the five-seed mean and one standard deviation.



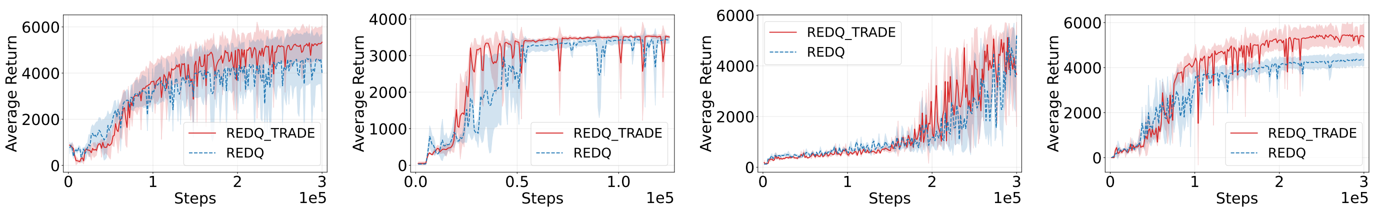
(a) *button-press-topdown* (b) *dial-turn* (c) *window-open*

Fig. 11. BRO comparisons on Meta-World. BRO+TRADE yields the clearest asymptotic improvement on *button-press-topdown*, stabilizing around 0.55–0.60 success compared with 0.30–0.35 for BRO, while improving or maintaining learning on *dial-turn* and *window-open*.



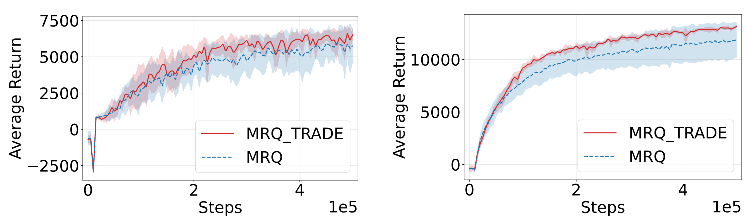
(a) *Ant* (b) *HalfCheetah* (c) *Hopper* (d) *Walker2d*

Fig. 12. Representative MuJoCo locomotion environments. Their different morphologies and contact dynamics provide additional high-dimensional tests of TRADE beyond the DMC benchmark.



(a) *Ant* (b) *Hopper* (c) *Humanoid* (d) *Walker2d*

Fig. 13. Additional REDQ comparisons on MuJoCo locomotion tasks. REDQ+TRADE generally reaches higher returns on *Ant*, *Hopper*, and *Walker2d* while remaining competitive on *Humanoid*, supporting adaptive refresh for ensemble critics.



(a) *Ant* (b) *HalfCheetah*

Fig. 14. Additional MR.Q comparisons on MuJoCo locomotion tasks. MR.Q+TRADE learns faster and reaches higher mean returns on both *Ant* and *HalfCheetah*, extending the benefit to a strong representation-learning baseline.